Jorge Loría

PhD in Statistics

Jorge Loría /loriaJ /jorge-loría jorge.loria@aalto.fi loriaj.github.io/

Academic Employment

Aalto University June 2024–Present
Postdoctoral Researcher, Advisor Dr. Samuel Kaski *Espoo, Finland*

Education

Purdue University 2019 – 2024
PhD Statistics, Advisor: Dr. Anindya Bhadra *Indiana, USA*

Thesis title: *On the use of α-stable random variables in Bayesian bridge regression, neural networks and kernel processes*

Universidad de Costa Rica 2015 – 2019
BS Pure Mathematics *San José, Costa Rica*

Universidad de Costa Rica 2014 – 2017
BS Actuarial Science *San José, Costa Rica*

Research Interests

- Heavy tails
- High dimensional data
- Bayesian inference
- Gaussian processes
- Computational methods
- Causal inference

Publications

Preprints

- Mäkinen L., **Loría, J.**, and Kaski, S. (2026+), Bayesian Meta-Learning with Expert Feedback for Task-Shift Adaptation through Causal Embeddings *(submitted)* [arxiv][github]
- Fernández-Loría, C. and **Loría, J.** (2026+), Inferring Effect Ordering Without Causal Effect Estimation *(submitted)* [arxiv]

Publications

- Björkman, Z., **Loría, J.**, Wharrie, S., and Kaski, S. (2026), Incorporating Expert Knowledge into Bayesian Causal Discovery of Mixtures of Directed Acyclic Graphs *(accepted to AISTATS 2026)* [arxiv][github]
- **Loría, J.** and Bhadra, A. (2025), Deep Kernel Posterior Learning under Infinite Variance Prior Weights. *The 13th International Conference on Learning Representations (ICLR 2025).* [link] [github]
- **Loría, J.** and Bhadra A. (2024). Posterior Inference on Shallow Infinitely Wide Bayesian Neural Networks under Weights with Unbounded Variance. *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*, **PMLR 244** 2331-2349 [link] [github]
- **Loría, J.** and Bhadra, A. (2024). SURE-tuned bridge regression. *Statistics and Computing* **34**, 30. [doi link] [github]
- Víquez, J. J., Víquez, J., Campos, A., **Loría, J.**, Mendoza, L. A. (2018). Demographic Modeling via 3-dimensional Markov Chains *Revista de matemática Teoría y aplicaciones* available in https://revistas.ucr.ac.cr/index.php/matematica/article/view/33608

Awards

Somos Purdue Award Class of 2024
recognition of commitment to excellence and for leading with *ganas* at Purdue University.

Bravo Award Spring 2024
one time staff award

Ross-Lynn Research Scholar Grant Spring 2024
provides funding for notable graduate students

Fully funded assistance to workshop on Bayesian Statistics and Statistical Learning December 2023
IMSI, Illinois, USA

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Department of Statistics Travel Award Summer 2023

Department of Statistics Teaching Academy Associate Nominee Spring 2023

Emily and Paul Kidwell graduate student excellence Award Spring 2022
given to graduate students in the Department of Statistics in Purdue University interested in careers in academia.

Bob and Marjorie McLean Scholarship Summers 2020 and 2021
given to distinguished graduate students of the Department of Statistics in Purdue University.

Bronze Medal 2015
awarded in Competencia Iberoamericana Interuniversitaria de Matemáticas

Honorable Mention 2014
awarded in Competencia Iberoamericana Interuniversitaria de Matemáticas

Talks and Posters Presented

Incorporating Expert Knowledge into Bayesian Causal Discovery of Mixtures of Directed Acyclic Graphs May, 2026
Poster in Artificial Intelligence in Statistics *Morocco*

Deep Kernel Posterior Learning under Infinite Variance Prior Weights April, 2025
Poster in International Conference on Learning Representations *Singapore*

Posterior Inference on Shallow Infinitely Wide Bayesian Neural Networks under Weights with Unbounded Variance July, 2024
Poster in Uncertainty in Artifical Intelligence *Barcelona, Spain*

Predicción en redes neuronales Bayesianas infinitas con pesos de varianza infinita May 22nd, 2024
Seminario Intersede *CIMAT, Guanajuato, México*

SURE-tuned Bridge Regression August 7th, 2023
Contributed Paper in Joint Statistical Meeting *Toronto, Canada*

SURE-tuned Bridge Regression July 10th, 2023
Contributed talk in XVI Latin American Congress of Probability and Mathematical Statistics *São Paulo, Brazil*

SURE-tuned Bridge Regression May 26th, 2023
Poster in Statistics and Optimization in Data Science Workshop, in Purdue Business School *Indiana, USA*

SURE-tuned Bridge Regression April 2023
Invited talk in a weekly colloquium at CIMPA Universidad de Costa Rica. *Online*

Learner Culture and Perceptions of Social Media Applications and Learning Management Systems for Language Learning April 2023
Paper session American Educational Research Association (AERA) *Illinois, USA*

Examining Learner Culture, Country, Gender, and English Public Speaking and Peer Feedback Self-Efficacy April 2023
Round table in AERA *Illinois, USA*

Learning the Ranking of Causal Effects with Confounded Data October 2022
Flash talk in INFORMS *Indiana, USA*

Research Experience

Purdue University

- Research Assistant to Dr Anindya Bhadra Spring 2022 and Fall 2023

» in the project Inference in Infinite Width Neural Networks with Infinite Variance

- Research Assistant to Dr Arman Sabbaghi Fall 2020-Summer 2021

» designing experiments and evaluating results for refinement of nanotellurene thickness

- Team leader in Statistics for the Community 2019-2021

» investigated the effects of having a bicycle to enable access to charity locations for impoverished people, with the LifeCycle program *(voluntary)*.

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Universidad de Costa Rica

- Research Assistant to Dr Luis Barboza 2016

» implementing an actuarial model for a projection model of Caja Costarricense del Seguro Social

- Research Assistant to Dr Juan José Víquez 2015

» implementing a multivariate Markov chains model for the projection of public universities in Costa Rica

Service to the Profession

PhD. Thesis advisor

» Lotta Mäkinen (ongoing, with S. Kaski as supervisor)

MSc. Thesis advisor

» Zachris Björkman (2024), Lotta Mäkinen (2025), and Axel Hedman (2025)

Reviewer

» Statistics in Medicine (2023), Neurips (2023), UAI (2024), ICLR (2024, 2026), AISTATS (2026)

Teaching Experience

Aalto University Fall 2022
Instructor *Espoo, Finland*

» Co-organized a seminar on causal inference, following Pearl's 'primer' book on causal inference.

Purdue University Fall 2022
Instructor *Indiana, United States*

» Taught one section of the course *Introduction to Statistics*. An introductory course for undergraduate students of computer science, mathematics, actuarial science, and statistics majors. Responsibilities included three weekly lectures, preparation of the relevant material, grading, and collaboration with the other instructors in the evaluation and design of the assignments of the course.

Purdue University Fall 2021 – Summer 2022, Summer 2023
Instructor *Indiana, United States*

» Taught two sections of the course *Elementary Statistical Methods*. An undergraduate level course for non-stem majors. Responsibilities included two weekly lectures to each section, preparation of the material, grading, leading teaching assistants and collaboration with other instructors for the evaluation of the course assignments.

Purdue University Fall 2019 – Spring 2021
Teaching Assistant *Indiana, United States*

» Held laboratories for two sections, office hours, and grading for the course *Elementary Statistical Methods*.

Universidad de Costa Rica Fall of 2017 & 2018
Instructor *San José, Costa Rica*

» Taught undergraduate course on basic functionalities of Excel and R. Responsibilities included lesson planning, course scheduling, and preparation of homework, exams, and course projects.

Industry Experience

Banco Promérica January 2019 – July 2019
Financial Risk Analyst, analyzed financial data to evaluate and forecast credit risk *San José Costa Rica*

Quanto Soluciones S.A. January 2017 – July 2018
Actuarial Model Programmer, implemented credit risk models *San José Costa Rica*

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Communication

Spanish *Native* **English** *Advanced* **Portuguese** *Basic*

Leadership Roles

- President of Statistics Graduate Student Organization 2021-2022 *Indiana, USA*

- Vice-president of Costa Rican Students Association at Purdue (CRISaP) 2021-2024 *Indiana, USA*